DOOSAN

Doosan Bobcat India Pvt Ltd, Bangalore

ENGINEERING TEST PROCEDURE

Subject: TEST SPEC PARAMETERS FOR C185WKUB-T2 (FAMILY 46824326)	Procedure Number:	46801805/I
	Revision Level:	B
	Date:	22-04-2021
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1.0 Purpose :

1.1 This document shows the C185 testing procedure for reference and to record the **testing performance, battery voltages, Air flow parameters (CFM) & Chassis light test** into the given tabular columns.

2.0 Fluid fill information :

2.1 Below listed fluids to be filled prior starting the machine. Refer to Test Spec 46826125 on the BOM

2.1.1 70652102 – Diesel Fuel – 13L
2.1.2 36899714 – Lubricant, Pro-Tec – 13 L
2.1.3 70646260 – Petronas Urania supremo 15W40-C14 Engine oil – 9.5 L
2.1.4 46830145 – Fleet guard EG premix 50/50 – 12L

2.2 For commissioning, add 4 liters of compressor oil. Suitable locations are: Oil supply connections, the airend air inlet, Vacuum port OR separator tank scavenge port.

3.0 Ratings :

Item	Rated Value	Acceptable Range
Safety valve setting:	200 psig	Valve to be marked with this rating
Minimum pressure:	75 psig	70 to 88 psig
Idle speed:	1700 RPM	1675 to 1725 RPM
Full load speed:	2550 RPM	2525 to 2575 RPM (See below)
Rated pressure:	100 psig	98 to 102 psig
Airflow:	185 CFM	175.8 to 194.3 CFM @ 2550 RPM rated speed

4.0 Special Instruction

4.1 Adjust full engine speed at beginning of test (while engine is cold) to maximum throttle position measuring crank shaft speed.
4.2 Set pressure regulator to 100 PSI @ 2550 rpm. Regulation pilot pressure to unloading should be 0 +/-1 @ 2550 rpm.
4.3 Run machine for 30 mins
4.4 At the end of the 30-minute test, recheck engine speed and adjust to within acceptable range if necessary.
4.5 If engine speed cannot be adjusted to a minimum of 2550 RPM, add 19 litres of fuel, run unit 5 minutes, and recheck speed.
4.6 If minimum speed of 2550 RPM is not achieved, report failure to QA.

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5.0 Setting up:

5.1 Position as level as possible. The design of these units permits a 15-degree sidewise limit on out-of-level operation.

5.2 When the unit is to be operated out-of-level, it is important:

5.2.1 To keep the engine crankcase oil level near the high-level mark (with the unit level).

5.2.2 To have the compressor oil level gauge show no more than mid-scale. Do not overfill either the engine crankcase or the compressor lubricating oil system.

5.2.3 The side doors must be closed to maintain a cooling air path and to avoid recirculation hot air.

6.0 Before Starting

6.1 Open service valve(s) to ensure pressure is relieved in receiver-separator system.

6.2 Close valve(s) in order to build up full air pressure and ensure proper oil circulation.

6.3 Check battery for proper connections and condition.

6.4 Check engine coolant level.

6.5 Check the engine oil level. Maintain per marks on dipstick.

6.6 Check the fuel level.

6.7 Check the compressor lubricating fluid level between bottom and midway of the sight glass on the separator tank.

7.0 Starting:

7.1 Move the POWER switch to ON.

7.2 Move Power switch to START position to crank engine. Hold switch in START position for approximately 5 seconds after engine starts.

7.3 Release Power Switch (it will automatically move to the ON position) when the engine starts and sustains running. Stop the unit within thirty (30) seconds of firing by moving the power switch off. Allow the unit to blow down normally. Check for leaks in all the joints and fittings. If a leak occurs, determine the cause and correct.

7.4 Then start the engine and allow to warm up 3 minutes.

7.5 Press the Service Air Button. Open air service valve(s) & set the machine performance parameters as given in 10th point.

7.6 Ensure the pointer in the pressure guage is over zero & Record the reading only while the canopy is in closed condition.

8.0 Stopping:

8.1 Close air service valves.

8.2 Allow the unit to run at idle for 3 to 5 minutes to reduce the engine temperatures.

8.3 Move Power Switch to OFF position.

8.4 When the engine stops, automatic blowdown valve should relieve system air pressure. If automatic blow-down valve malfunction is suspected, open manual blowdown valve

9.0 Safety shut down:

9.1 Engine coolant temperature: 110°C.

9.2 Low engine oil pressure

9.3 Compressor Air discharge temperature: 120°C

Revision History

REV	DESCRIPTION OF CHANGE	DATE	ORIGINATOR	APPROVER
A	ORIGINAL RELEASE	16-10-2020	Nimalan R	Ajay RB
B	LOW IDLE RPM RANGE REVISED FROM 1650-1750 TO 1675-1725 RPM AND LOW IDLE REGUALTION PRESSURE RANGE REVISED FROM 40-45 TO 43-49 PSIG	22-04-2021	Satish/Girish	Ajay RB

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10. Machine details :

Machine Serial Number	000305 BLAF J01
Engine Serial Number	7MN B237
Airend Serial Number	S21DA00-01087
Diesel Filled (Inhouse Testing Purpose)	30 ltr

11. Battery voltage check :

Initial battery voltage		Starting voltage (During Cranking)		Charging alternator	
Acceptance Criteria	Tested value	Acceptance Criteria	Tested value	Acceptance Criteria	Tested value
11.5V-12.7V	12.70 V	9V-11V	10.69 V.	13V-14.5V	14.39V

12. Testing parameters:

Machine Running hours	Test date
2 hours	26/12/21

Sl.No	Testing condition	Time	Engine RPM		Air discharge pressure ADP (psig)		Reg. Pressure (psig) (Check by connecting the pressure gauge on the regulation airline)	
		Mins	Accptence Criteria (RPM)	Tested (RPM)	Acceptance Criteria (Psig)	Tested Pressure (Psig)	Acceptance Criteria (Psig)	Tested Pressure (Psig)
1	Flushing (High idle)	20	2525-2575	2561	70-88	75	0	0
2		25		2560		75		0
3	Rated (Full load)	45	2525-2575	2547	98-102	100	0-1	0
4		50		2546		100		0
5	Unloading (Low idle)	60	1675 - 1725	1706	118-122	120	43-50	45

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13. Air flow parameters & Calculated capacity :

Nozzle Size 0.50 inch	Reference speed 2550 RPM (Rated)	Note: -While testing airflow, canopy should be closed. -Parameters should be recorded at rated condition. -Machine has to be run for atleast 20 to 30 minutes before taking values.
Nozzle Sr.No: 40792		

S.No	Nozzle pressure Psi	Nozzle temperature °C	Barometric pressure bar	Relative humudity %	Actual speed Rpm	Air discharge pressure Psi	Ambient temperature °C	Air intake temperature °C
	35	71.7°C	0.911	46.4%	2546	100	29.5°C	29.9°C

CALC. CAPACITY 184.8 CFM

Tested By : M. NARASIMHA Date: 26/12/21

13. Chassis Light Test :

Test	Running Lights	Brake Lights	Turn Lights
Pass	✓	✓	✓
Fail			

Tested By : V. RAJKUMAR Date: 26/12/21

Tested By	Production Engineer	Quality Engineer
M. NARASIMHA SRIRAM KUMAR. V	[signature]	[signature]